Comparative Topic-Model Analysis of iGEM and the Synthetic Biology Literature

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Background & Objective

Synthetic biology spans a growing body of **academic research** and a worldwide community of student innovators competing in the **iGEM** (International Genetically Engineered Machine) competition.

Despite shared goals, these two communities evolve on different timelines and emphasize different topics.

Objective

Map both corpora onto a **shared topic landscape** to identify:

- Where the two communities **converge** or **diverge**
- Which community addresses a topic **first** (temporal precedence)

Data & Methods

Data sources

Corpus	Records
OpenAlex papers	24,202
iGEM team projects	4,548

Pipeline

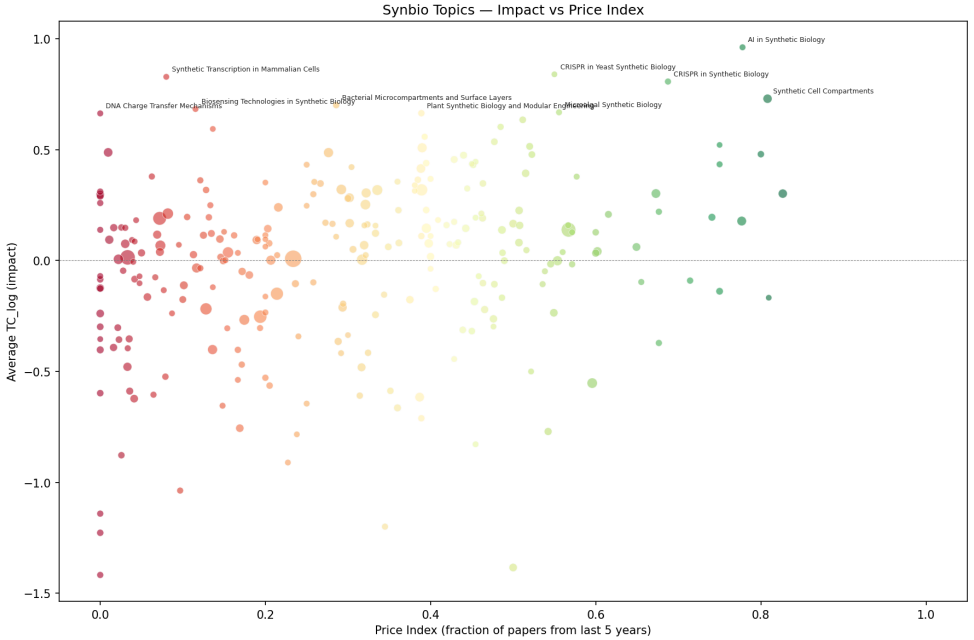
```
1. Collect    → OpenAlex API + iGEM registry
      ↓
2. Embed      → Sentence-Transformers
      ↓
3. Cluster    → BERTopic (UMAP + HDBSCAN)
      ↓
4. Analyze    → Hierarchy · Overlap · Precedence
```

Results — Papers Topic Overview

24,202 papers clustered into:

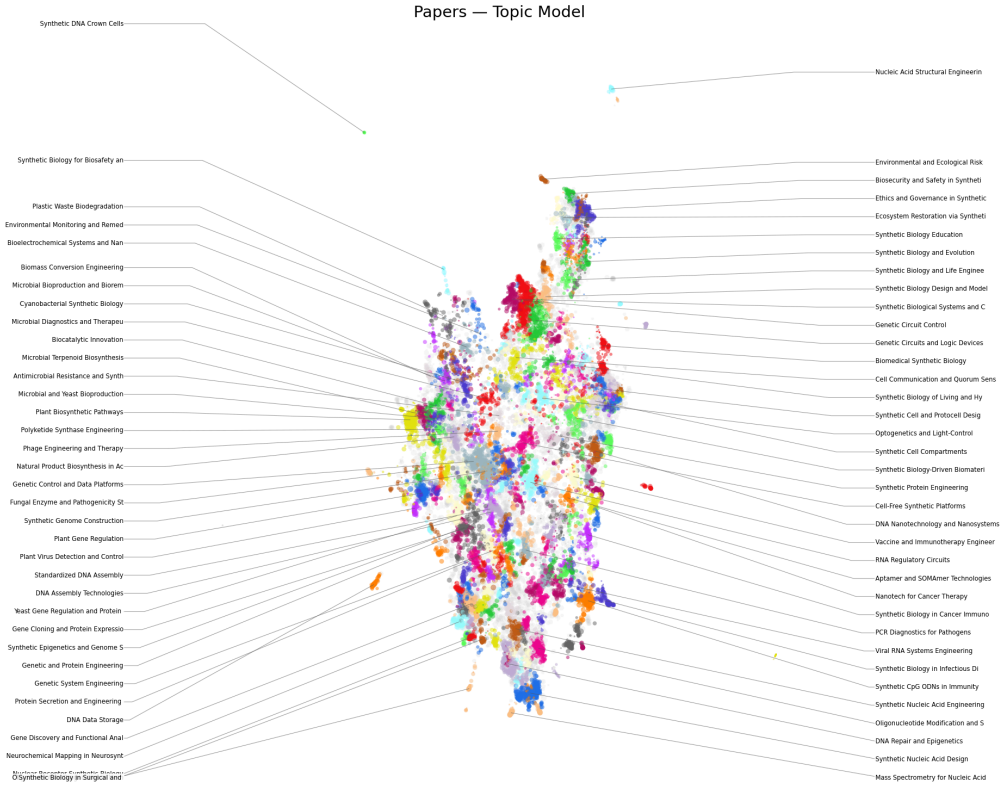
- **4** high-level groups
- **44** mid-level groups
- **263** low-level topics

Each dot is a topic. X = **Price Index** (recent-paper fraction), Y = **citation impact** (log-normalized).



Results — Papers Topic Map

2-D UMAP projection of **24,202** papers, colored by topic.

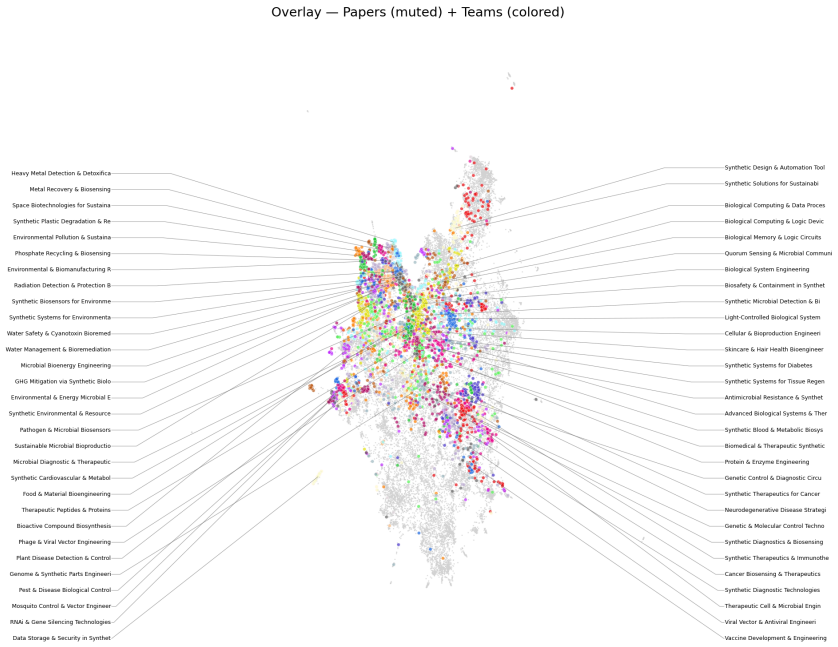


Results — iGEM Teams & Overlap

4,548 iGEM projects clustered into:

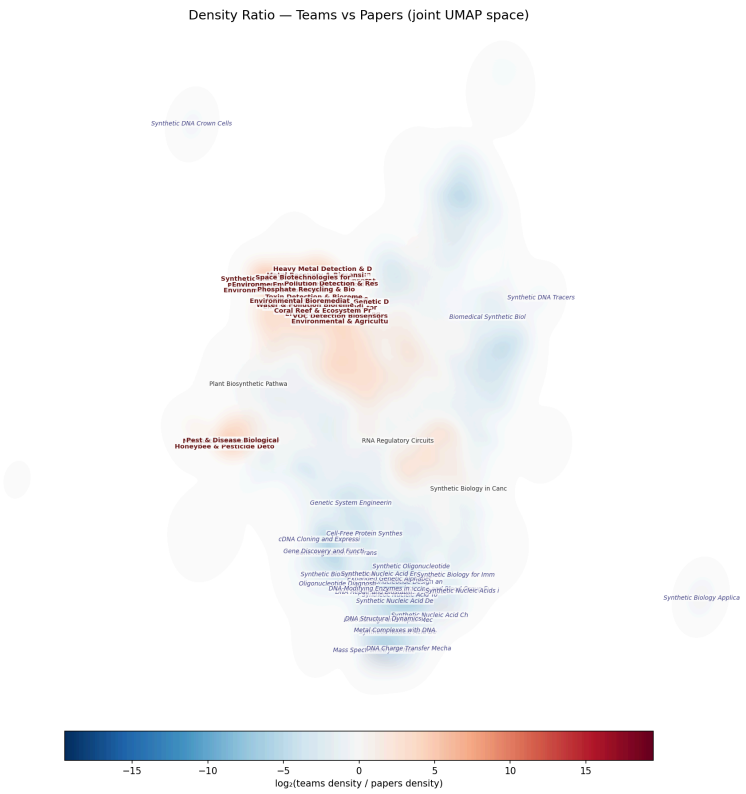
- **10** high-level groups
- **25** mid-level groups
- **161** low-level topics

Papers in grey; team projects in color. Joint UMAP space enables direct spatial comparison.



Results — Density Heatmap

Red = iGEM-dominated, **blue** = literature-dominated, **white** = balanced.



Results — Temporal Precedence

Literature precedes iGEM

#	Topic	Δ yr
1	Synbio in Infectious Disease	+15.5
2	Cytokine Gene Regulation	+13.5
3	Fungal Enzyme & Pathogenicity	+10.1
4	Viral Synbio & Vaccines	+8.7
5	Vascular Gene & Receptor Eng.	+8.7
6	High-Throughput Genetic Eng.	+7.1
7	Synbio-Driven Biomaterials	+6.1
8	Vaccine & Immunotherapy	+5.7
9	Pathogen Detection via Synbio	+5.6
10	Plant Gene Regulation	+5.4

iGEM precedes Literature

#	Topic	Δ yr
1	AI in Synthetic Biology	-6.0
2	Microbial Chassis Dev.	-4.5
3	CRISPR in Synbio	-4.5
4	AI-Driven Imaging & Data	-4.4
5	Microalgal Synbio	-4.1
6	CRISPR Control Technologies	-3.6
7	Microbial Community Eng.	-3.2
8	Plant Resilience Eng.	-3.2
9	CRISPR-Based Bacterial Reg.	-3.0
10	Synth. Promoter Eng. in Yeast	-3.0

Δ yr = avg year (teams) – avg year (papers). Top 10 Δ yr each.